

Kerr-lens mode-locked bidirectional dual-comb ring laser for broadband dual-comb spectroscopy

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Received 31 March 2016; revised 15 June 2016; accepted 20 June 2016 (Doc. ID 262310); published 7 July 2016

Fourier-transform spectroscopy is an indispensable tool for analyzing chemical samples in scientific research as well as the chemical and pharmaceutical industries. Recently, its measurement speed, sensitivity, and precision have been shown to be significantly enhanced by using dual-frequency combs. Moreover, recent demonstrations of inducing nonlinear effects with ultrashort pulses have enriched the utility of dual-comb spectroscopy. However, wide acceptance of this technique is hindered by its requirement for two frequency combs and active stabilization of the combs. Here, we overcome this predicament with a Kerr-lens mode-locked bidirectional ring femtosecond-pulse laser that generates two broadband frequency combs with slightly different pulse repetition rates and a tunable yet highly stable rate difference. Since these combs are produced by one and the same laser cavity, their relative coherence stays passively stable without the need for active stabilization. To show its utility, we demonstrate broadband dual-comb spectroscopy with the single laser. © 2016 Optical Society of America

OCIS codes: (140.3560) Lasers, ring; (140.4050) Mode-locked lasers; (190.3270) Kerr effect; (300.6300) Spectroscopy, Fourier transforms; (300.6530) Spectroscopy, ultrafast.

<http://dx.doi.org/10.1364/OPTICA.3.000748>

1. INTRODUCTION

Since it was commercialized more than 40 years ago, the Fourier-transform spectrometer based on Michelson-based Fourier-transform spectroscopy [1], most notably known as a form of Fourier-transform infrared spectroscopy, has long been used as a user-friendly, robust, and reliable instrument for analyzing and identifying chemical compounds in diverse fields such as molecular precision spectroscopy [2], organic synthesis, chemical catalysis [3], biological analysis [4,5], environmental monitoring, and clinical pathology [6,7]. Over the last decade, it has been proven that its measurement speed, sensitivity, and precision can significantly be enhanced [8–23] by using laser frequency combs [24]. This technique, known as dual-comb spectroscopy, is based on the principle (analogous to the principle of a sampling oscilloscope) that a probe light composed of two frequency combs with slightly different pulse repetition rates works as a mechanical scan-less temporal interferometer due to their linearly increasing relative time delay between the pulse pairs. In the frequency domain, it can be understood as a multi-heterodyne detection of the pairs of neighboring comb lines that have slightly different line spacings, by which the optical frequency combs are down-converted to a single radio-frequency comb. This scan-less interferometer has been demonstrated to outperform the traditional

mechanically driven Michelson-based Fourier-transform spectrometer by several orders of magnitude in terms of measurement time, sensitivity, and spectral resolution (hence precision) [23]. Furthermore, it has recently been proven that the dual-comb system based on ultrashort pulses enables nonlinear dual-comb spectroscopy, such as coherent Raman dual-comb spectroscopy [13,15], two-photon excitation dual-comb spectroscopy [21], and saturated absorption dual-comb spectroscopy [16]. The realization of the nonlinear dual-comb spectroscopy has expanded the utility of dual-comb spectroscopy to high-speed biomedical imaging and precision spectroscopy with ultrahigh spectral resolution.

Unfortunately, despite its excellent capabilities, dual-comb spectroscopy has not been adopted beyond research laboratories. This is due to its costly requirement for two laser frequency combs and comb stabilization systems, such as f - $2f$ interferometers or external reference cavities with feedback control circuits. Since mode-locked lasers (free-running combs) have independent fluctuations in their pulse repetition rate and carrier-envelope phase, their dual-comb interferogram is significantly distorted, spoiling its Fourier transform (spectrum) with chromatic artifacts. For this reason, a frequency stabilization system is typically required to tightly phase lock one frequency comb against the other comb

and hence to satisfy the requirement for the high relative coherence between the combs in order to fully benefit from the superior performance of dual-comb spectroscopy. However, such high-precision comb-stabilization techniques are only available at optical metrology laboratories with considerable expertise and infrastructure. Consequently, the high cost and complexity of the dual-comb spectrometer betray the virtue of the simple and handy Fourier-transform spectrometer for which it has been widely used by diverse researchers for the last 40 years.

In this paper, to overcome this difficulty, we propose and demonstrate a surprisingly simple yet robust method for dual-comb spectroscopy based on a bidirectional ring laser. The laser is composed of a free-running Kerr-lens mode-locked laser cavity that produces two broadband laser frequency combs with slightly different pulse repetition rates and a tunable yet highly stable rate difference. This lasing principle, which we refer to as direction-dependent lasing due to the asymmetrical Kerr nonlinearity, is based on our revisit to and further development of the effect that there exists a slight difference in cavity round-trip time between the two counter-propagating mode-locked pulses—an effect that is contrary to the photonics community's traditional understanding that the two laser outputs should have the same pulse repetition rate, as they share the same optical cavity. Since these combs are produced by one and the same laser cavity, their relative coherence stays passively stable. As a result, it enables highly stable broadband dual-comb spectroscopy without the need for any mechanical moving parts and complicated comb stabilization systems with feedback control circuits. Moreover, since the laser is based on the Kerr lens mode locking with a Ti:sapphire crystal, it generates frequency combs that consist of extremely short pulses. This source of the ultrashort and broadband combs promises its versatile utilization for dual-comb spectroscopy in both the linear and nonlinear regimes. As a proof-of-principle demonstration, we show broadband dual-comb spectroscopy with the single free-running laser. This simple and robust dual-comb laser is expected to be highly effective for *in situ* spectroscopy in a diverse range of fields.

2. RESULTS

A. Principle of the Bidirectional Dual-Comb Ring Laser

As schematically shown in Fig. 1, the bidirectional dual-comb ring laser is based on a four-mirror bowtie ring cavity with a Ti:sapphire crystal pumped by a diode-pumped solid-state laser at 532 nm with an average power of approximately 9 W (a similar design can be found in Ref. [25]). All the cavity mirrors, including two concave mirrors (M1, M2) and a convex mirror (M3), and an output coupler (OC) are dielectric coated to provide negative group-delay dispersion for intra-cavity pulse compression. The ring cavity generates ultrashort pulse trains by means of soft-aperture Kerr-lens mode locking [26] in either the uni-directional or the bi-directional lasing mode [27]. The alignment procedure is no different from that of the conventional Kerr-lens mode-locked laser. Our manual adjustment of the cavity mirror (M2) stochastically triggers mode locking either in a uni- or bi-directional manner. We expect that a systematic fine adjustment with an electrically driven actuator can improve its reproducibility. In the case of bidirectional mode locking, the repetition rates of the two outputs can be made identical or slightly different, depending on the cavity alignment. As shown below, we have

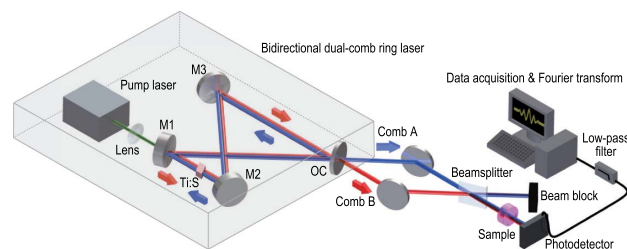


Fig. 1. Schematic of the bidirectional dual-comb ring laser and dual-comb spectroscopy system. The laser is based on a four-mirror bowtie ring cavity with a Ti:sapphire (Ti:S) crystal pumped by a diode-pumped solid-state laser at 532 nm. All the cavity mirrors, including two concave mirrors (M1, M2) and a convex mirror (M3), and an output coupler (OC) are dielectric coated to provide negative group-delay dispersion for intra-cavity pulse compression. The ring cavity generates ultrashort pulse trains by means of soft-aperture Kerr-lens mode locking in either the uni-directional or the bi-directional lasing mode. In the case of bidirectional mode locking, the repetition rates of the two outputs can be made identical or slightly different, depending on the cavity alignment. The two outputs of the laser (Comb A, Comb B), spatially combined by a beam splitter, pass through the sample and are incident onto the photodetector. The low pass-filtered photodetector signal is digitized and Fourier transformed for dual-comb spectroscopy.

observed stable bidirectional mode locking without the need for a hard aperture for the negative feedback, which was required for stable bidirectional mode locking in Ref. [27]. The origin of the difference in repetition rates between the two pulse trains can be explained by their slight difference in round-trip time due to the direction-dependent Kerr nonlinearity in the laser crystal, namely, the self-steepening effect [28,29]. The self-steepening effect varies the group velocity in the laser crystal, depending on the optical intensity. Specifically, in our laser cavity, there are mainly three factors that cause the direction-dependent optical intensity in the laser crystal: (1) a non-uniform intensity distribution of the pump laser in the laser crystal along the beam direction, (2) a slight offset of the laser crystal from the middle point between the cavity mirrors (M1, M2), and (3) bidirectional Kerr lensing due to the first two factors. Based on this theory, we have estimated the degree of the difference in repetition rate and verified that it agrees well with our experimental observation (see Supplement 1 for details).

B. Characteristics of the Two Outputs of the Dual-Comb Laser

The spectra of the two outputs of the bidirectional dual-comb ring laser are shown in Fig. 2(a). They have nearly identical spectral shapes, with a center frequency of 358 THz and an FWHM bandwidth of 26 THz (corresponding to 61 nm in wavelength). The slight difference in shape between the two spectra indicates non-identical lasing between the counter-propagating modes in the cavity. The broad bandwidth (26 THz) of the laser is essential for broadband dual-comb spectroscopy and superior to previously reported single-oscillator-based mode-locked dual-comb lasers based on a fiber laser [30,31], a semiconductor disk laser [32], or a monolithic cavity laser [33], which only provide narrow bandwidths (less than 300 GHz). The output pulses are slightly chirped, but can be compressed down to 12 fs by an external dispersion compensator. This extreme ultrashort pulse generation

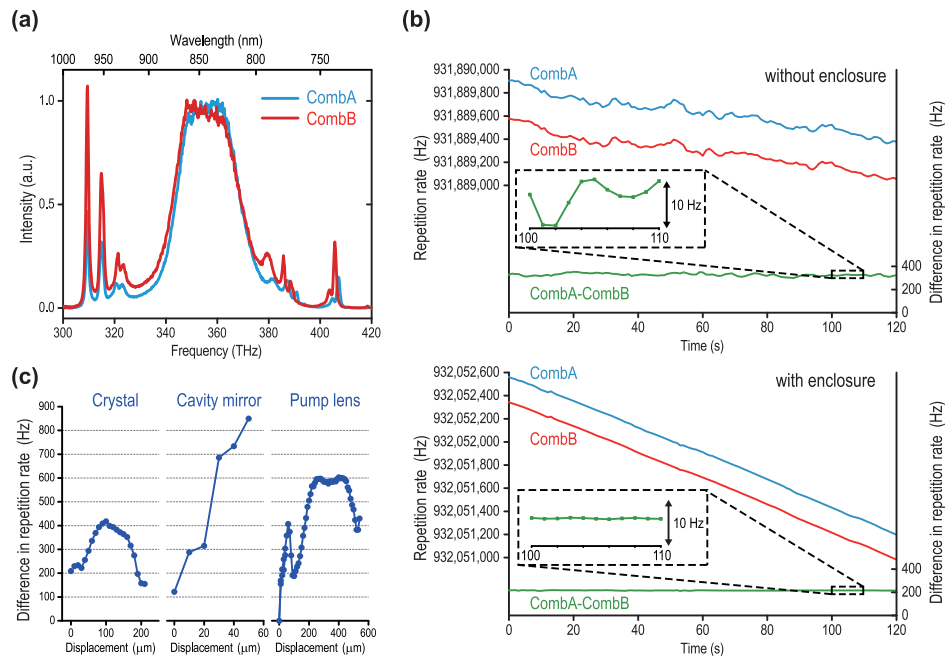


Fig. 2. Basic performance of the bidirectional dual-comb ring laser. (a) Spectra of the two laser outputs (shown in blue and red) measured by a grating-based spectrometer. (b) Temporal behavior of the pulse repetition rates of the two combs (shown in blue and red) and the difference between them (shown in green). The insets show the zoomed difference in pulse repetition rate over 10 s. (c) Tunability of the difference in pulse repetition rates between the two laser outputs as a function of the displacement of the Ti:sapphire crystal, the cavity mirror M2, and the focusing lens for the pump laser with respect to the initial positions where the laser has been mode locked. The positive directions of the displacements (x-axis) in the figure are defined as the forward, forward, and backward directions of the pump laser's propagation for the crystal, mirror, and lens, respectively.

directly from the oscillator is the unique outcome of the Kerr-lens mode-locked Ti:sapphire laser, which cannot be realized with any other types of mode-locked lasers. The two combs have an intensity profile of TEM₀₀ and average optical powers of 340 and 310 mW, respectively. With simple modifications, the dual-comb laser can be configured to produce the spectrally broadest (up to 175 THz) and temporally shortest (down to 5.4 fs) pulses directly from the oscillator [34] with the lowest noise level [35].

C. Stability of the Dual-Comb Laser in the Pulse Repetition Rate Difference

The pulse repetition rate of each laser output and the difference in pulse repetition rates between them are shown in Fig. 2(b). While each repetition rate fluctuates around 932 MHz due to environmental disturbances (e.g., air flow, ambient temperature fluctuations, and mechanical vibrations), the difference between them remains highly stable at a quasi-constant value of 325 Hz with a standard deviation of 3.4 Hz over minutes. This common-mode noise of the two repetition rates is a significant advantage for dual-comb spectroscopy, in which the difference in pulse repetition rates between the two combs must be kept constant during the measurement of interferograms. The repetition rate difference can further be made more stable by, for example, enclosing the laser cavity with a simple housing made of aluminum [Fig. 2(b)]. The figure indicates a drastic improvement in the stability, resulting in a standard deviation of 0.1 Hz around a repetition rate difference of 216 Hz. We anticipate that a hermetically sealed cavity, commonly employed for commercial products, can further stabilize the repetition rate difference against air flow and ambient temperature fluctuations.

D. Tunability of the Dual-Comb Laser in the Pulse Repetition Rate Difference

The difference in pulse repetition rate between the two laser outputs can be tuned over ~1 kHz by multiple techniques. As shown in Fig. 2(c), it can be varied by displacing either the Ti:sapphire crystal, the cavity mirror (M2), or the focusing lens for the pump laser with the laser pulses mode locked during the displacement. While the mirror displacement has a high frequency tunability of 15 Hz/μm over a large frequency range, the crystal and lens displacements have a relatively low tunability of 3 Hz/μm (evaluated in the quasi-linear region) and work in a narrower frequency range. These features can be taken advantage of for high-speed modulation of the repetition rate difference using the mirror displacement [9] and for precise tuning of the difference using the crystal and lens displacements.

E. Relative Coherence between the Two Outputs of the Dual-Comb Laser

The relative temporal coherence between the two pulse trains is directly evaluated by measuring the linewidth of the radio-frequency beat note between a pair of comb lines from the two combs (see Section 5). As shown in Fig. 3(a), the beat note has an FWHM linewidth of 13 kHz over a measurement duration of 1 ms, indicating that the interferogram measured by the dual-comb spectrometer can maintain coherence for 77 μs. In addition, as shown in Fig. 3(b), the beat note is kept stable within 200 kHz over 60 min, indicating high long-term stability in the relative coherence between the two pulse trains. This long-term stability firmly supports continuous measurements for a long duration of time without suffering from a detrimental aliasing effect

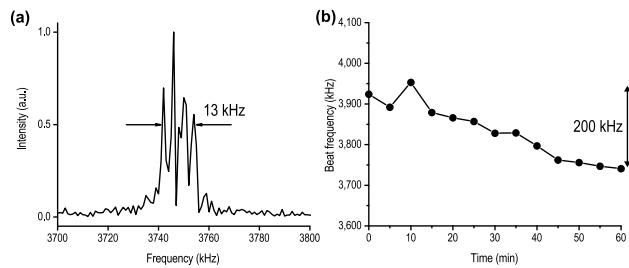


Fig. 3. Relative coherence between the two combs. (a) Beat note between a pair of comb lines from the two combs measured in 1 ms, showing high relative coherence between them. (b) Frequency of the beat note sampled at every 5 min over 60 min, showing high long-term stability in the relative coherence between the two combs.

that would otherwise occur in the case of unstable coherence. Note that these results show the coherence property of the combs around the laser's carrier frequency where carrier envelope offset frequency noise (CEO noise) has little influence on the combs.

F. Broadband Spectroscopy with the Dual-Comb Laser

Finally, we performed dual-comb spectroscopy with the bidirectional dual-comb ring laser. Figure 4(a) shows a sequence of dual-comb interferograms acquired by detecting two spatially overlapped frequency combs with a repetition frequency difference of 270 Hz (without a sample). Fourier transforming each interferogram over 20 μ s around each burst point produces a broadband dual-comb spectrum that can repetitively be acquired every 3.7 ms [Fig. 4(b)]. We then conducted dual-comb absorption spectroscopy of an Nd:YVO₄ crystal with a spectral resolution of 93 GHz and an acquisition duration of 67 μ s over a spectral range of 18 THz (corresponding to 40 nm) centered at 367 THz (corresponding to 817 nm) (Fig. 5). Here, we acquired spectra with and without the sample and plotted the transmittance spectrum. The dual-comb spectrum clearly shows the absorption lines of Stark-split sub-levels from the ground level (⁴I_{9/2}) to the ²H_{9/2} and ⁴F_{5/2} bands of neodymium ion (Nd³⁺)

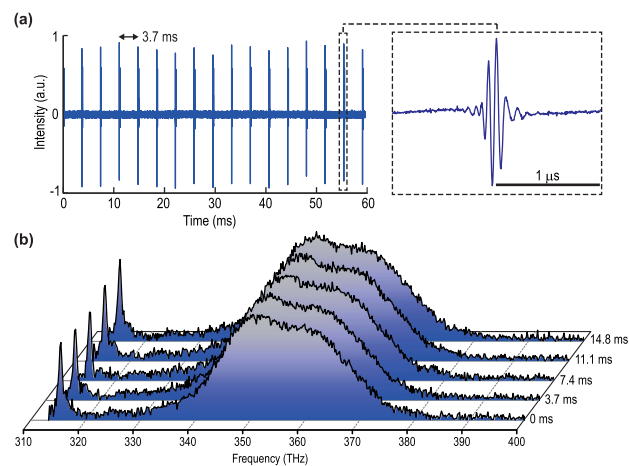


Fig. 4. Dual-comb spectroscopy with the bidirectional dual-comb ring laser. (a) Continuously measured signal that includes multiple interferograms. The interval between the consecutive interferograms is 3.7 ms. The inset shows one of the interferograms. (b) Consecutive broadband dual-comb spectra that are obtained by Fourier transforming each interferogram over 20 μ s around each burst point.

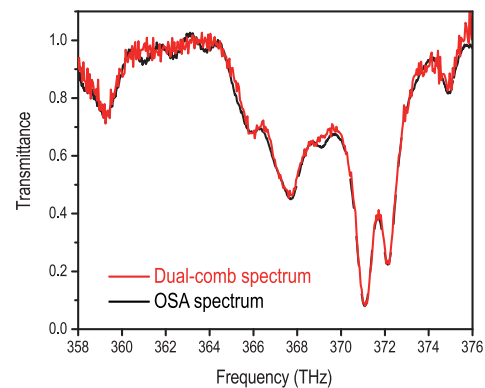


Fig. 5. Dual-comb absorption spectroscopy of a Nd:YVO₄ crystal with the bidirectional dual-comb ring laser. The dual-comb absorption spectrum of a Nd:YVO₄ crystal was obtained in the transmission mode (shown in red) with a spectral resolution of 93 GHz and an acquisition duration of 67 μ s over a spectral range of 18 THz (corresponding to 40 nm) centered at 367 THz (corresponding to 817 nm). The spectrum agrees well with the spectrum of the same sample obtained by a conventional grating-based optical spectrum analyzer (black).

doped into yttrium orthovanadate (YVO₄). The spectrum is in good agreement with the spectrum of the same sample obtained by a conventional grating-based optical spectrum analyzer. Note that the spectral resolution of 93 GHz is high enough to measure molecular vibrations in a liquid phase whose spectral linewidth is $\sim$ 100 GHz. Therefore, our proof-of-principle system is applicable for high-speed spectroscopic measurements of biological samples that are mostly in the liquid phase.

3. DISCUSSION

The capabilities of the dual-comb spectrometer with the dual-comb laser can further be enhanced by incorporating a few techniques into it or extending it beyond what has been demonstrated in this paper. First, phase error-correction techniques, such as real-time adaptive sampling [13] and real-time computational re-sampling [12], can easily be implemented on the system in order to achieve higher spectral resolutions, while a careful phase noise analysis of the system is required to estimate the level of achievable precision. To achieve high accuracy, however, the combs must be stabilized to an optical or rf reference, hence referenced to an absolute frequency standard. Second, while in our proof-of-principle demonstration, we have performed spectroscopy in the near-infrared region (750–950 nm), the dual-comb laser can be extended to other spectral regions, such as XUV, UV, visible, mid-infrared, and terahertz via nonlinear frequency conversion or spectral broadening. It is guaranteed by our laser's ability to produce ultrashort pulses. When the spectra are extended to a region far from the carrier frequency of the combs via a process such as nonlinear spectral broadening, the extended CEO noise can degrade the mutual coherence between the combs. However, a nonlinear frequency conversion process such as difference frequency generation or optical rectification generates frequency-converted combs that have similar sensitivities to the CEO noise as the fundamental combs. Therefore, such a frequency conversion to a region far from the carrier frequency can be accomplished without critically degrading the mutual coherence. Third, the dual-comb laser can also be used for other dual-comb

applications than dual-comb spectroscopy, such as asynchronous optical sampling [36], absolute long-range distance measurements [37], and optical coherence tomography [38]. Finally, by virtue of its ultrashort pulses, the dual-comb laser can directly be employed for nonlinear spectroscopy, such as coherent Raman scattering spectroscopy [13,15], two-photon excitation spectroscopy [21], and saturated absorption spectroscopy [16].

4. CONCLUSIONS

In summary, we have developed the Kerr-lens mode-locked bidirectional dual-comb ring laser emitting two femtosecond-pulse trains with slightly different repetition rates and a tunable yet highly stable rate difference. The ultrashort pulse trains enable versatile dual-comb spectroscopy in both the linear and nonlinear regimes. Since the dual-comb pulses share the common-mode fluctuations, their temporal interferometry is passively stable without the need for active stabilization. Furthermore, we have experimentally shown the passively stable coherence between the combs. Our proof-of-principle demonstration of dual-comb spectroscopy with the dual-comb laser shows its ability to acquire broadband absorption spectra at a high scan rate of ~ 1 ms. This high scan rate, combined with the laser's long-term robust operation, can bring us opportunities not only to investigate fast transient dynamics (e.g., protein folding kinetics and plasma physics), but also to conduct many measurements in a short duration of time (e.g., flow cytometry and wide-field scanning microscopy). This free-running dual-comb operation with the single laser is expected to be a solution to simple and robust dual-comb spectroscopy beyond research laboratories and hence effective for field use in diverse areas.

5. METHODS

A. Design of the Dual-Comb Laser

The Ti:sapphire mode-locked laser is based on a bowtie ring cavity that consists of four mirrors, including two concave mirrors (M1, M2) with a radius of 30 mm, a convex mirror (M3) with a radius of 1000 mm, and a flat output coupler (OC). All of the mirrors are dielectric coated to achieve high reflectivity in a broad spectral range: $>99.9\%$ (620–1050 nm) for the concave mirrors, $>99.8\%$ (670–1000 nm) for the convex mirror, and 99% (700–900 nm) for the output coupler. The mirror coatings are also designed to add negative dispersion with -50 , -60 , and <20 fs² for the concave mirrors, the convex mirror, and the output coupler, respectively. A Ti:sapphire crystal with a thickness of 2.3 mm is placed at a point near the middle point between M1 and M2 at a Brewster angle against the beam path. The crystal is pumped by a CW DPSS green laser at 532 nm (Millenia eV, Spectra Physics) with an average power of 8–9 W focused by a plano-convex lens with a focal length of 40 mm.

B. Evaluation of the Dual-Comb Laser

The spectra of the laser outputs shown in Fig. 2(a) were measured by a grating-based spectrometer with a resolution of 400 GHz (LSM-Mini, Ocean Photonics). The repetition rates shown in Figs. 2(b) and 2(c) were measured in a gate time duration of 1 s with two frequency counters (FCA3003, Tektronix) referenced to a 10 MHz rubidium frequency standard (FS725, Stanford Research Systems).

C. Measurement of the Beat Note between a Pair of Comb Lines from the Two Outputs of the Dual-Comb Laser

For the beat note measurement, a CW external-cavity diode laser running at 790 nm (TLK-L780M, Thorlabs) was used as an intermediate. First, the beat note between the CW laser and one of the combs was generated by a fast photodetector, from which the beat note at the lowest frequency was extracted by a low-pass filter. The beat note between the CW laser and the other comb was also generated in the same way. Next, the beat notes were mixed by an electronic mixer, from which only the subtraction signal was extracted by filtering. Finally, the signal was digitized by a 14-bit data acquisition board (ATS9440, Alazartech) at a sampling rate of 20 MS/s. The Fourier transformation of the sampled data yields the beat note shown in Fig. 3(a). To produce Fig. 3(b), the beat-note measurements were performed every 5 min.

D. Dual-Comb Spectroscopy with the Dual-Comb Laser

For dual-comb spectroscopy, the two outputs were combined by a 50/50 beam splitter into a dual-comb beam, which is incident onto an amplified high-speed Si photodetector. The detector signal was low pass filtered at a cut-off frequency of less than half the repetition rate of the combs (<466 MHz) and digitized by a 14-bit data acquisition board. For the absorption spectroscopy measurement shown in Fig. 5, an Nd:YVO₄ crystal was placed as a sample in between the beam splitter and the photodetector. A short-pass filter with a cut-off wavelength of 850 nm was placed in front of the photodetector.

Funding. ImPACT Program of the Council for Science, Technology and Innovation (Cabinet Office, Government of Japan); Burroughs Wellcome Fund (BWF); Sumitomo Foundation; Konica Minolta Imaging Science Foundation.

Acknowledgment. We are grateful to K. Yoshioka for loaning us the optical components.

See [Supplement 1](#) for supporting content.

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